ICS, SCADA, and IIoT Vulnerabilities

CompTIA PenTest+ (PT0-003)
Domain: 4.0: Attacks and Exploits
Objective: 4.9: Explain common attacks against specialized systems.

Video Overview

This video provides an overview of Industrial Control Systems (ICS), Supervisory Control and Data Acquisition (SCADA) systems, and the Industrial Internet of Things (IIoT). You will learn about their significance in various industries and how their unique vulnerabilities can be exploited. This material is designed to help you prepare for the PenTest+ exam by understanding specialized system attacks and their real-world implications, as well as to equip you with foundational knowledge for a career in penetration testing focusing on operational technology environments.

Key Topics Covered

- **Operational Technology (OT) Terminology**: Defining OT as the overarching category for industrial automation and critical infrastructure systems.
- **Industrial Control Systems (ICS)**: Technologies that directly control physical processes, such as robots and assembly lines.
- **SCADA Systems**: Systems designed for remote monitoring and control across multiple, widespread operations.
- **Industrial Internet of Things (IIoT)**: The integration of smart devices and sensors in industrial environments, combining OT with IT.
- **Connectivity and Pivoting Risks**: How connecting OT systems to IT networks and the internet creates pathways for attackers.
- **Legacy Systems and Outdated Software**: The security challenges posed by older, unsupported systems that cannot be easily updated.
- **Common Vulnerabilities**: Weak authentication, poor network segmentation, and insufficient encryption in ICS/SCADA/IIoT environments.
- **Real-World Exploits (e.g., Ripple20)**: Examples of significant vulnerabilities affecting various industrial and consumer devices.
- **Threats to OT**: Malware, Denial of Service (DoS/DDoS), Human Machine Interface (HMI) attacks, protocol-specific attacks (CAN bus, Modbus), human error, insider threats, and side-channel attacks.

- **Radio Frequency (RF) Connected Controller Attacks**: Exploiting wireless communications in industrial settings.

Student Notes Section